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# GSTGAT: Gated spatiotemporal graph attention network for traffic demand forecasting

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**Abstract**

Urban traffic demand forecasting is an important component in the implementation of intelligent transport systems (ITS). Urban traffic demand data is a spatiotemporal data, describing the amount of traffic demand generated by different areas or stations within a city along the time dimension. Although there has been considerable research work, researchers still face several challenges in predicting accurately, including the capture of hidden features in the temporal dimension of such spatiotemporal data, and the capture of dynamic dependent changes in the spatial dimension. These are even more difficult for long-time series prediction tasks. This paper designs a multivariate temporal forecasting model specifically adapted to traffic demand to address these challenges, called the Gated Spatiotemporal Graph Attention Network (GSTGAT). GSTGAT is based on the Transformer framework and the whole model is used in an end-to-end manner. First of all, it uses the gated self-attention to extract temporal features in the sequence. Secondly, graph attention is used to capture the spatial dependencies among different variables in the unstructured space. Finally, the use of gated recurrent units in combination with hidden spatial states is proposed to capture multiple levels of spatial dependencies. Experimental results on the taxi dataset in New York and the bicycle dataset in San Francisco Bay Area show that the authors' proposed model outperforms other state-of-the-art models and improves the prediction accuracy.

## 1 | INTRODUCTION

In recent years, with the rapid development of urban transport and the increasing demand for transport, more and more cities are committed to developing Intelligent Transport Systems (ITS). This phenomenon has led many scholars to research related issues, such as urban traffic signal timing optimization, rail route planning and personalized travel route recommendations. One of the fundamental issues that needs to be addressed is traffic demand forecasting. By anticipating the amount of traffic demand in various areas of the city, traffic management departments can allocate traffic resources reasonably in advance, thus alleviating traffic congestion, saving travellers' travel time and avoiding the waste of public resources. In order to provide better services for people's daily travels and maximize the utilization of transport resources, it is necessary to forecast the demand.

Over the years, researchers have worked hard on traffic demand forecasting. Urban transport demand data is a type of spatiotemporal data generated over time in different areas of the city. In the early days, when the amount of available data and the power of computing devices were insufficient, the prediction methods adopted by researchers could be divided into two approaches: one is based on traditional parametric models, and the other is based on traditional machine learning models. The former mainly includes Historical Average (HA) [1], Autoregressive Integrated Moving Average model (ARIMA) [2] and Vector Autoregression (VAR) [3]. These methods mainly make assumptions of linear dependence on time series, whereas the variation of traffic data along time is often non-linear and complex. As a result, they usually perform poorly in practical applications. Then, some subsequent machine learning methods appear, which mainly contain Support Vector Regression (SVR) [4], Hidden Markov Model (HMM) [5] and Extreme

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Gradient Boosting (XGBoost) [6] etc. Although these methods relieved the linearity assumption and achieved better results than the previous methods, their performance often relies on feature engineering, which is time-consuming and labour-intensive to produce features by hand. And these methods take more account of temporal correlation and ignore spatial correlation.

Later, with the widespread application of Convolutional Neural Network (CNN) [7] and Recurrent Neural Network (RNN) [8] in the fields of image processing and Natural Language Processing (NLP) respectively, more and more research has started to apply deep neural networks to traffic prediction. These research efforts can be divided into approaches based on CNNs [9, 10], RNNs [11, 12], and fusing CNNs and RNNs [13–15]. Compared with the previous traditional parametric models and traditional machine learning models, these deep learning-based methods can achieve better results without an over-reliance on manual labor. While these methods can achieve some success, there are still issues about accurate and long-term forecasting that need to be addressed. It is well known that urban traffic demand data is a type of spatiotemporal data that describes the amount of traffic demand generated by different areas or stations within a city along the time dimension. We need to solve how to model this spatiotemporal data considering both the temporal and spatial dimensions.

For the feature extraction in the temporal dimension, existing research work has either used a RNN approach or a one-dimensional CNN approach. The work based on RNNs uses Long Short-Term Memory (LSTM) [16] or Gated Recurrent Unit (GRU) [17] to capture temporal correlations. But RNNs are susceptible to the vanishing gradient problem when capturing temporal correlations, which is particularly severe in long-term prediction problems [18]. The work based on CNNs uses a one-dimensional convolutional approach, where the convolutional kernel has a limited receptive field, and therefore it can only observe dependencies in the temporal sequence from a local perspective. For feature extraction in the spatial dimension, existing research work has either used a CNN approach or a Graph Convolutional Networks (GCN) approach. The researchers attempted to grid the entire traffic study area, using CNNs to extract spatial correlations. However, due to the road network's topology, the sites are often in a non-Euclidean spatial relationship with each other. Traditional CNNs are unable to extract features correctly on such unstructured data, that is, traditional discrete convolution cannot maintain translation invariance on non-Euclidean structured data. Later, GCN have also been widely used in transportation in recent years due to their excellent performance in processing non-Euclidean structured data [19], which can establish topological associations and thus extract spatial features efficiently [20–23]. However, in graph convolution operations, it often requires expensive matrix operations and depends on the structural information of the graph, so it is difficult to achieve satisfactory results.

To address the above problems, we propose a new deep neural network learning framework called Gated Spatiotemporal Graph Attention Network (GSTGAT). The contributions of this work are summarized as follows:

- We propose using a multi-head gated self-attention mechanism to capture temporal features in traffic data, solving the vanishing gradient problem in long-time series modelling tasks.
- We have designed a novel gated recurrent graph attention unit (GRGAU) based on graph attention and GRUs to efficiently capture spatial correlations in traffic data in a dynamic manner.

Extensive experiments are conducted on two real-world traffic datasets (taxi demand dataset and bicycle demand dataset) and our GSTGAT model is compared with other baseline approaches. The experimental results demonstrate that our proposed method outperforms other state-of-the-art models. To assess the effectiveness of GSTGAT, we also performed ablation experiments to evaluate the impact of each component of GSTGAT on performance.

The rest of the paper is organized as follows. We present some previous work related to traffic demand forecasting in Section 2. Section 3 presents some relevant definitions of the traffic demand forecasting problem. Section 4 explains the details of the model in this paper from three aspects. Section 5 provides detailed experiments on a publicly available dataset and a comparative analysis with other methods. Conclusions and future work are given in the final section of the paper.

## 2 | RELATED WORK

Due to the popularity of sensing technology and the Internet of Things (IoT), a large amount of traffic data is recorded and saved, allowing researchers to mine useful information from traffic data to predict future traffic. Moreira-Matias et al. [24] first used three time-series prediction methods (Time-Varying Poisson Model, Weighted Time-Varying Poisson Model and ARIMA Model) to predict the distribution of taxi demand in the short term, based on the Global Positioning System (GPS) data of taxis. Li et al. [25] proposed an improved ARIMA-based method to predict the spatiotemporal distribution of passengers in hotspot areas. The main work considered historical data and the periodicity of the number of pickup events to the hotspot. Wei et al. [26] proposed a zero-grid ensemble spatiotemporal model to predict the passenger demand for online taxis, and in particular, a specific zero-grid prediction module was designed for regions with zero demand. Tong et al. [27] proposed a unified linear regression model with high dimensional features (LinUOTD) to predict taxi demand. In particular, to train this model effectively, they used the distributed learning based on parameter servers and hash-based labelling methods. Wang et al. [28] proposed a data-driven prediction algorithm for bike-sharing demand forecasting. Their main work was to construct sparse demand and supplied matrices using a relatively small number of bicycle check-in and check-out data, and considered the correlation between regional points of interest and the use of inter-regional bicycle. Huang et al. [29] proposed a prediction algorithm based on a Gaussian Poisson distribution to predict the number of available shared bikes in each station,

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and they included an environmental parameter to measure the impact of environmental factors on-bike use. However, these methods based on mathematical statistics and machine learning often perform poorly in practical applications due to the considerable complexity of traffic demand data variability. These methods are unable to capture the hidden correlations of spatiotemporal data at the same time and are more dependent on feature engineering, which is often time and effort consuming to perform feature extraction, so the prediction effectiveness and applicability are very limited.

With the rise of deep learning techniques in areas such as Computer Vision (CV) and NLP, many deep learning-based methods have been used to tackle the problem of traffic demand prediction. Chang et al. [30] proposed a framework based on artificial immune systems and artificial neural networks to predict the demand for bike-sharing, where they introduced the concept of cells, antigens and antibodies in artificial immune systems for prediction. Wang et al. [31] designed an end-to-end neural network framework to predict taxi demand, where multiple data sources (weather, hailing orders etc.) were used to assist to improve prediction accuracy, but a small number of hand-crafted features were required. Xu et al. [11] proposed a method based on LSTM and Gaussian kernel to forecast urban taxi demand. Rodrigues et al. [12] fused text information and time-series information for regional taxi demand forecasting. In this they used word embedding, one-dimensional CNN and attention mechanisms to extract hidden features in text information and LSTM to extract temporal correlations in time-series data (historical demand counts, historical weather and additional event information). Jiang et al. [9] treated urban taxi demand data as image data form and used two-dimensional CNN and Residual Network (ResNet) to perform taxi demand prediction. However, the feature extraction was only performed for the local space and could not take into account the global space, and was not effective for the extraction of time-series features. Yao et al. [13] proposed a multi-view neural network model combining CNN and LSTM by simultaneously capturing spatiotemporal correlations for taxi demand prediction. They first extracted spatial correlations from the spatial views by CNN and then connected the contextual features at the corresponding time by LSTM. Zhou et al. [14] combined CNN and ConvLSTM [32] on an encoder–decoder framework and applied the attention mechanism to build urban taxi and bicycle demand forecasting models. Liu et al. [33] proposed a CSTN model to solve the taxi origin–destination demand prediction problem. In particular, the local spatial context (LSC) module was designed to extract local spatial features of taxi demand, essentially using convolutional kernels of different sizes for the feature extraction operation.

However, it is difficult to capture spatial correlations effectively through CNN approaches to unstructured data; researchers have gradually been modelling this kind of unstructured data with graphs. Davis et al. [34] proposed the use of a GraphLSTM (Graph-based LSTM) model, by representing the Voronoi spatial partitions as nodes on an arbitrarily structured graph. The existing spatiotemporal forecasting models are mostly designed based on GCN. Yu et al. [10] designed a spatiotemporal GCN model based on GCN and gated temporal convolutional networks to perform traffic prediction. Chai et al. [15] proposed a multi-graph CNN model to predict bike-sharing demand. Specifically, they used three heterogeneous inter-station graphs (distance graph, interaction graph and correlation graph) of the bike-sharing system for separate graph convolution and refusion to extract spatial correlation, and combined LSTM on the basis of an encoder–decoder framework to extract temporal correlation. Although the work applying graph convolution has made significant breakthroughs in traffic demand forecasting, these research works using graph convolution for spatial feature extraction rely heavily on the inter-station location relationships, and sometimes inter-station adjacency matrices are often not easily accessible. Later, Wu et al. [23] proposed to automatically learn an adaptive adjacency matrix from training data for using in graph convolution operations to capture hidden spatial correlations in spatiotemporal data, and combined with dilation convolution to design a spatiotemporal prediction model for application to traffic speed prediction. Wu et al. [20] proposed a graph learning module for using in graph convolution operations to extract dependencies among multiple variables, and combined it with temporal convolution to design a spatiotemporal prediction model for traffic speed prediction. Ke et al. [35] proposed the Spatio-Temporal Encoder–Decoder Residual Multi-Graph Convolutional network (ST-ED-RMGC), a novel deep learning model for predicting ride-sourcing demand of various origin-destination (OD) pairs. The model constructs OD graphs, which utilize adjacent matrices to characterize the non-Euclidean pair-wise geographical and semantic correlations among different OD pairs.

However, even though some research work applying graph convolution has proposed methods for adaptive graph structure generation, it is difficult to achieve satisfactory results, and graph convolution operations often require expensive matrix operations.

## 3 | PROBLEM DEFINITION

In this section, we will briefly introduce the relevant definitions of the traffic demand forecasting problem.

- Station-based transport.

The modes of transport can be divided into two categories: stationed transport and non-stationed transport. For example, bus and metro trips are stationed transport and taxi and private car trips are non-station transport. For non-stationed transport, passengers are usually clustered in certain places, although the locations of their origins or destinations are discrete. For example, a shopping centre with many taxi orders at the front door could be similar to a station at this location. We can convert non-stationed transport into stationed transport in some ways, for example, by clustering the pick-up and drop-off locations into different zones, for each zone representing a station.

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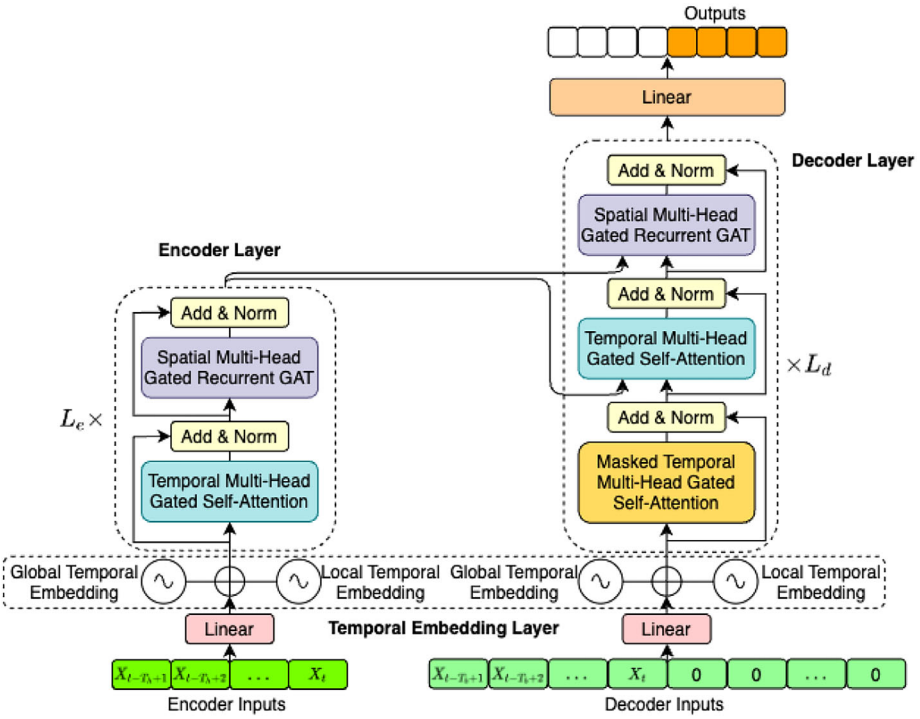



**FIGURE 1** The GSTGAT model is based on the encoder–decoder architecture and consists of three parts: the temporal embedding layer, the $L_e$ encoder layer and the $L_d$ decoder layer. GSTGAT, gated spatiotemporal graph attention network.

- Traffic demand feature matrix.

For a particular mode of transport, the demand for the station-based transport can be divided into two types: the demand for pick-ups and the demand for drop-offs, which can be expressed in terms of the number of pick-ups and drop-offs at a station respectively. We define the connectivity of stations in a transport network as a undirected graph $G = (V, E)$, where V is the set of all stations and $E$ is the set of all connected lines between any two stations. At time step $t$, graph $G$ has the traffic demand feature matrix $X_t \in \mathbb{R}^{N \times f}$, where $N$ denotes the number of stations and $f$ denotes the traffic demand dimensions (in GSTGAT it is 2, 2 indicates pick-up and drop-off demand, or 1 when the demand for pick-up or drop-off is forecasted separately.).

- Traffic demand forecasting.

At time step $t \in \{1, 2, ...\}$ given the traffic demand feature matrix $X = \{X_{t-T_h+1}, X_{t-T_h+2}, \dots, X_t\}$ for the past $T_h$ time steps, the objective is to obtain the traffic demand feature matrix $\widehat{Y} = \{X_{t+1}, X_{t+2}, \dots, X_{t+T_p}\}$ for the future $T_p$ time steps.

## 4 | METHODOLOGY

In this section, we will present the details of GSTGAT. Figure 1 shows its overall composition. The model is built on the basis of the Transformer [36] framework, while a temporal multi-head gated self-attention module and a spatial multi-head gated recurrent graph attention module are designed to extract temporal and spatial features in the temporal and spatial dimensions, respectively. GSTGAT consists of a temporal embedding module, an encoder and a decoder, where the encoder and the decoder are composed using the same layer stacking approach, and use residual connectivity [37] and layer normalization [38] to ensure that the model can still be trained effectively as the layers increase.

The workflow of GSTGAT is as follows. First, the original input of the encoder is $X = \{X_{t-T_h+1}, X_{t-T_h+2}, \dots, X_t\}$ and the original input of the decoder is $\{X_{t-T_h+1}, X_{t-T_h+2}, \dots, X_t, 0, 0, \dots, 0\}$, which are the traffic demand feature matrix at $T_h$ and $T_h + T_p$ time steps respectively. $T_h < T_p$, both of which are mapped from the feature dimension from $f$ to the high-dimensional $f'$ through the linear layer. The embedded input of the encoder and the embedded input of the decoder are obtained by the temporal embedding layer respectively. Then, the encoder obtains an output $X^{(L_e)}$ containing spatiotemporal information and a hidden state $H_{T_h}^{(L_e)}$ containing spatial information of the current time step through the $L_e$ encoder layer starting with the initial value $X^{(0)}$ layer by layer. Combined with the final outputs $X^{(L_e)}$ and $H_{T_h}^{(L_e)}$ of the encoder, the decoder obtains $\widehat{Y}^{(L_d)}$ through the $L_d$ layer decoder layer with the initial value $\widehat{Y}^{(0)}$ layer by layer. Finally, $\widehat{Y}^{(L_d)}$ passes through the linear layer and maps the feature dimension from $f'$ back to $f$, and the traffic demand feature matrix at the end $T_p$ time steps is intercepted to obtain the final prediction result.

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## 4.1 | Temporal embedding

RNN structures such as LSTM [16] and GRU [17] can capture time-series patterns through their recurrent structures and usually do not need to rely on the timestamp of the data. But the Transformer architecture uses only a self-attention mechanism for capturing temporal dependencies. Although Transformer can avoid the RNN structure repetitively and speed up the training process, the self-attention mechanism does not perceive the position of each element in the sequence because each element in the sequence enters the encoder and decoder stacks at the same time. However, temporal information plays a crucial role in capturing temporal correlation. First, the information from adjacent time points is more informative. For example, the information at 10 AM is more informative for predicting traffic demand at 11 AM compared to traffic demand information at 7 AM. Second, the information from the same time points is more informative. For example, to forecast the traffic demand of this Monday, the information of last Monday is more informative than last Tuesday. Because of the regularity of human activity, the information has a certain periodicity at the same time point. The embedding of the above local and global time information can help our model to better capture the relevance of the time-series data.

For the embedding of local and global time information, the following solution is used. Assume that at time step $t$, the feature matrix $X_t$ is available and that information on dimensions such as minutes, hours, days, weeks and months belonging to that time can be obtained from time step $t$. The global time feature matrix $GE_t \in \mathbb{R}^{N\times f'}$ for traffic demand at time $t$ can be obtained by projecting the dimensions to the $f'$ dimension through a linear layer, while the local time feature matrix was obtained using the location coding proposed by Vaswani et al. [36], as shown in Equation (1).

$$PE_{(\mathbf{pos},\mathbf{i})} = \begin{cases} \sin\left(\mathbf{pos}/(\mathbf{10000})^{2\mathbf{i}/\mathbf{f}'}\right), \mathbf{i\%2 = 0} \\ \cos\left(\mathbf{pos}/(\mathbf{10000})^{2\mathbf{i}/\mathbf{f}'}\right), \mathbf{i\%2 = 1} \end{cases} \tag{1}$$

where $pos \in \{1, 2, \ldots\}$ denotes the position located in the time series, and $i \in \{1, 2, \ldots, f'\}$ denotes the position located in the feature series. And the local time feature matrix $PE_t \in \mathbb{R}^{N\times f'}$ of the traffic demand at time step $t$ can be obtained by Equation (1). Then the input feature matrix $X_t^{(0)} \in \mathbb{R}^{N\times f'}$ contains the global and local time features embedded at time step $t$, which can be calculated by Equation (2) as

$$X_t^{(0)} = X_t + GE_t + PE_t \tag{2}$$

## 4.2 | Encoder

The encoder of the model consists of $L_e$ identical coding layers, and each layer consists of two basic modules, a temporal multi-head gated self-attention module and a spatial multi-head gated recurrent graph attention module, both of which capture features of the traffic demand data in the temporal dimension and the spatial dimension, respectively.

### 4.2.1 | Temporal multi-head gated self-attention

The computational flow of the multi-head gated self-attention module for time is shown in Figure 2 and mainly consists of a self-attention mechanism and a gating operation. A self-attention mechanism is a variant form of the attention mechanism [39], which reduces the reliance on external information and is better at capturing the internal relevance of the data. A common approach to implementing the self-attention mechanism is scaled dot product attention [36], which is defined as shown in Equation (3).

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{1}{\sqrt{f'}}QK^T\right)V \tag{3}$$

where Q, K, V and $f'$ represent the query, key, value and their dimensions, respectively. The key and value have the same dimension, while the query dimension can be different from the key and value, and in our model all three are two-dimensional feature matrices on the stations along the time dimension.

The multi-head self-attention mechanism allows the model to simultaneously focus on information from different representation subspaces at different locations [36]. In order to reduce the effect of weakly correlated information to model, we add an output gate [16] to control this by performing linear projection and sigmoid operations on the temporal data, thus enabling the model to learn a pattern to control the outcome of the output after the multi-head self-attention mechanism. Our temporal multi-head gated self-attention is defined as shown in Equation (4).

$$\begin{cases} head_i = \text{Attention}(Q_i, K_i, V_i) \odot \sigma(Q'_i) \\ \text{GatedSelfAttention} = \text{Concat}(head_1, \ldots, head_h)W^O \end{cases} \tag{4}$$

where $h$ denotes the number of heads, $d = f'/h$, $\odot$ is the Hadamard product, $Q_i = QW_i^Q$, $i \in \{1, 2, \ldots, h\}$, $K_i = KW_i^K$, $V_i = VW_i^V$, $Q'_i = QW_i^{Q'}$. The parameter matrices $W_i^Q$, $W_i^K$, $W_i^V$ and $W_i^{Q'}$ are $\mathbb{R}^{f'\times d}$, and $W^O$ is $\mathbb{R}^{hd\times f'}$. In the first layer of the encoder, the input feature matrix is $X^{(0)}$ and all nodes are calculated by the temporal multi-head gated self-attention to obtain the feature matrix $M^{(1)} \in \mathbb{R}^{N\times T_h\times f'}$, with the same data dimension before and after the calculation.

### 4.2.2 | Spatial multi-head gated recurrent GAT

The spatial multi-head gated recurrent graph attention network (GAT) is shown in Figure 3, which mainly consists of GAT [40] and GRU [17]. GAT can extend the traditional convolutional operation from structured data to graph-structured data and capture unstructured information hidden in the graph

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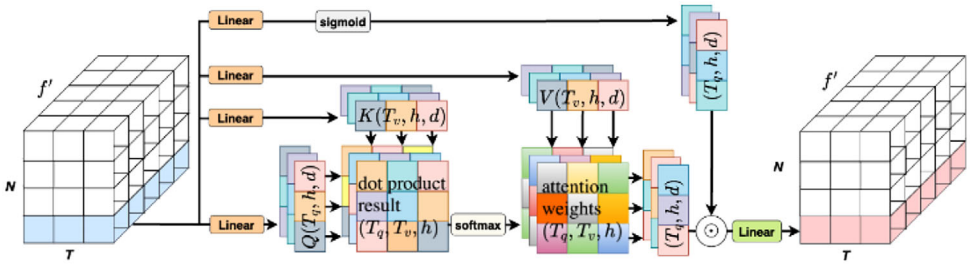



**FIGURE 2** Flowchart for calculating the multi-head gated self-attention.

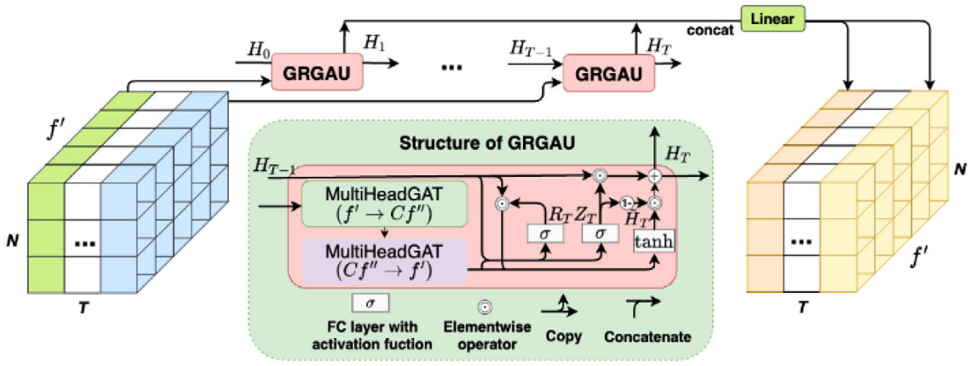



**FIGURE 3** Multi-headed gated recurrent graph attention structure.

just as well as GCN. The general idea of GAT is to use an attention network to aggregate information about nodes in the neighbourhood, that is, a sub-network is used to calculate the importance of neighbouring nodes to the target node. Aggregation is done on neighbouring nodes based on the calculated degrees of importance, while the sub-network is learnable, which enhances the generalization ability of the model. In our approach, given the input data $M_t = \{m_t^1, m_t^2, \dots, m_t^N\} \in \mathbb{R}^{N \times f'}$ at time step $t$, where $m_t^i \in \mathbb{R}^{f'} (i = 1, 2, \dots, N)$ denotes the feature information of node $i$ at time step $t$, the correlation between the target node $i$ and all neighbouring nodes $j (j = 1, 2, \dots, N)$ is calculated as in Equation (5).

$$\alpha_t^{ij} = softmax(LeakyReLU(W^a(W^m m_t^i \parallel W^m m_t^j))) \tag{5}$$

where $\parallel$ denotes the splicing operation and the parameter matrices $W^m \in \mathbb{R}^{f'' \times f'}$ $W^a \in \mathbb{R}^{2f''}$, $f''$ are the hidden feature dimensions. It is using multi-head attention to perform independent correlation calculations on the above equations, and then splicing their outputs together, as shown in the following equation:

$$\widetilde{m}_t^i = \|_{c=1}^{C} \sigma\left(\sum_{j \in N_j} \left(\alpha_t^{ij}\right)_c W_c^{\alpha} m_t^j\right) \in \mathbb{R}^{Cf''} \tag{6}$$

where $(\alpha_t^{ij})_c$ is the weight coefficient calculated for the group $c$ correlation and the parameter matrix $W_c^{\alpha} \in \mathbb{R}^{f'' \times f'}$, which can be calculated to obtain the feature matrix $\widetilde{m}_t^i \in \mathbb{R}^{Cf''}$ of node $i$ at time step $t$. Formally, the multi-head GAT can be defined as follows:

$$\text{MultiHeadGAT}(M_t, C) = \|_{i=1}^{N} \|_{c=1}^{C} \sigma\left(\sum_{j \in N_j} \left(\alpha_t^{ij}\right)_c W_c^{\alpha} m_t^j\right) \tag{7}$$

The above equation allows the calculation of the feature matrix $\widetilde{M}^t = \{\widetilde{m}_t^1, \widetilde{m}_t^2, \dots, \widetilde{m}_t^N\} \in \mathbb{R}^{N \times Cf''}$ for all nodes.

If graph attention is used to extract spatial features from the data at different time points, it is assumed that the spatial feature distributions at different time points are independent, which is not valid. Because the spatial feature distribution at one time step affects the spatial feature distribution at the next time step, hence, it is necessary to capture the spatial feature distribution at this time step by using the information of the spatial feature distribution at the previous time step. To capture this dynamic information across spatial dimensions, we further design a GRGAU, as shown in Figure 3, and the computational equation is defined as follows:

$$\begin{cases} X'_t = MultiHeadGAT(MultiHeadGAT(M_t, C), 1) \\ R_t = \sigma(X'_t W_{xr} + H_{t-1} W_{hr} + b_r) \\ Z_t = \sigma(X'_t W_{xz} + H_{t-1} W_{hz} + b_z) \\ \tilde{H}_t = \tanh(X' W_{xh} + (R \odot H_{t-1}) W_{hh} + b_h) \\ H_t = Z_t \odot H_{t-1} + (1 - Z_t) \odot \tilde{H}_t \end{cases} \tag{8}$$

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where $M_t \in \mathbb{R}^{N\times f'}$ and $H_t \in \mathbb{R}^{N\times f''}$ are the input and output feature matrices respectively, $R_t$ is the reset gate, $Z_t$ is the update gate and the parameter matrices $W_{xr}$, $W_{x\tilde{z}}$,$W_{xh}$, $W_{hr}$, $W_{h\tilde{z}}$ and $W_{hh}$ are $\mathbb{R}^{f''\times f''}$, $b_r$, $b_{\tilde{z}}$ and $b_h$ are $\mathbb{R}^{f''}$.

In the first layer of the encoder, the feature matrix calculated by the temporal multi-head gated self-attention module is input to the $T_h$ GRGAU unit to calculate the output. The output results are combined and the features are mapped from $f''$ dimension back to $f'$ dimension by a linear layer. Finally, we get an output $X^{(1)} \in \mathbb{R}^{N\times T_h\times f'}$ with spatiotemporal information and a hidden state $H_{T_h}^{(1)} \in \mathbb{R}^{N\times f''}$ with spatial information of the current time step.$X^{(L_e)}$ and $H_{T_h}^{(L_e)}$ can be obtained after layer-by-layer computation through the $L_e$ encoder layer.

## 4.3 | Decoder

The decoder uses generative inference decoding as proposed by Zhou et al. [41] to generate the output sequence, consisting of $L_d$ identical decoder layers. Each layer consists of three basic modules, two temporal multi-head gated self-attention modules and a spatial multi-head gated recurrent graph attention module. The temporal multi-head gated self-attention module differs from the encoder. In that module, the first module captures the decoder input sequences and uses a masked mechanism to prevent each temporal location from focusing on future sequence information, and the second module is used to capture the correlation between the decoder sequence (query) and the encoder output sequence $X^{(L_e)}$(key and value). The spatial multi-head gated recurrent graph attention module also differs from that in the encoder in that the initial hidden state input is the last hidden state $H_{T_h}^{(L_e)}$ by the encoder.

After the decoder$L_d$ layer, the output $\widehat{Y}^{(L_d)}$ can be obtained. And then the linear layer maps the dimension $f'$ back to the $f$, intercepting the traffic demand feature matrix at the end $T_p$ time steps to obtain the final forecast result $\widehat{Y} \in \mathbb{R}^{N\times T_p\times f}$.

# 5 | EXPERIMENT

In this section, we present an experimental comparison with other methods on real-world traffic dataset and conclude with an ablation analysis to verify the effectiveness of each module.

## 5.1 | Datasets

The taxi dataset and the bicycle dataset were chosen for this paper, considering that the taxi and bicycle types of travel meet the point-to-point transport needs of urban residents. We summarize the information about the dataset in Table 1. In addition, the dataset is described in detail as follows:

(1) NYC Taxi[1]: This dataset consists of 13.7 million taxi trip records for New York City from 1 November 1 2019 to 31 December 2019, and contains the following key information: pick-up area number, drop-off area number, pick-up time and drop-off time. Four transit-intensive areas of Manhattan are selected for the study (Lower Town, Midtown, Upper East Side, Upper West Side). There are 51 zones, each of which serves as a virtual station, and we count the pick-up and drop-off demand for each zone at 15-min intervals. The specific data-processing process is as follows: for the taxi order data records of these 2 months, the data records with missing key values (tpep_pickup_datetime, tpep_dropoff_datetime, PULocationID, DOLocationID, passenger_count) are first removed, and then, the numbers of picking up records and dropping off records are counted by each area id according to 15 min as time interval, and finally, the NYC Taxi dataset in this study can be obtained. The NYC Taxi dataset has a data dimension of 51 × 5856 × 2.

(2) SF Bay Area Bike[2]: This dataset consists of over 70,000 bike-share rides in the San Francisco Bay Area from 1 September 2013 to 31 October 2013, and contains the following key information: ride start station, ride end station, start time and end time. Seventy stations are included, and we count the pick-up and drop-off demand for each station at 30-min intervals. The specific data-processing process is as follows: for the bike order data records of these 2 months, the data records with missing key values (start_date, start_station_id, end_date, end_station_id) are first deleted, and then the numbers of picking up and dropping off records are counted at 30-min intervals for each station id, and finally the SF Bay Area Bike dataset in this study is obtained, which has a data dimension of 70 × 2928 × 2.

**TABLE 1** Dataset statistics.

| Datasets | Nodes | Time range | Sample rate |
|---|---|---|---|
| NYC Taxi | 51 | 11/012019–12/31/2019 | 15 min |
| SF (San Francisco) Bay Area Bike | 70 | 09/01/2013–10/31/2013 | 30 min |

## 5.2 | Evaluation metrics

We use Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) to assess the predictive performance of different methods. Both are defined as follows:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\widehat{y}_i - y_i| \tag{9}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\widehat{y}_i - y_i)^2} \tag{10}$$

[1] https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page

[2] https://www.kaggle.com/benhamner/sf-bay-area-bike-share

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where $n$ is the total number of samples tested, $y_i$ and $\widehat{y}_i$ represent the true and predicted values of sample $i$, respectively.

## 5.3 | Baselines

- HA: We calculate the average of historical demand values to predict future values.
- SVR [4]: a method that uses linear support vector machines to perform regression tasks.
- XGBoost [6]: a method based on gradient boosting trees.
- FC-LSTM [16]: a method that combines LSTM and feedforward neural networks.
- LSTNet [42]: a method that combines CNN with RNN.
- Graph WaveNet [23]: a spatiotemporal GCN that combines diffusion graph convolution with dilation convolution.
- Multivariate time-series graph neural networks (MTGNN) [20]: a graph neural network framework suitable for multivariate time series prediction, which designs specialized graph learning modules to extract dependencies between variables.

## 5.4 | Experimental settings

All datasets[3] are divided into training, validation and test sets in the ratio of 7:1:2 and use the standard normalization method. The historical time step $T_h$ is set to 12, the prediction time step $T_p$ is set to 12 and the $T_b$ steps before prediction are set to $T_h/2$. We used the pytorch[4] framework to implement the GSTGAT model with the learning rate set to 0.001 and the batch size set to 32. All the methods are optimized by the Adam [43] algorithm, using Mean Square Error (MSE) as the loss function. The Early Stopping [42] method is used to prevent overfitting, as the patience is set to 5 and the training epochs are no more than 100. For all the compared methods, we adjust their key parameters to ensure optimal performance.

## 5.5 | Results and discussion

All of our experiments are averaged over five training repetitions, and Table 2 shows the results of our method compared to other methods at prediction 12 time steps. Our method achieves the best performance results on both datasets. Since the demand for taxis in New York City is much greater compared to the demand for bicycles in the San Francisco Bay Area, the evaluation metrics calculated by the various methods are larger in value on the taxi dataset than on the bicycle dataset.

HA, SVR, XGBoost and FC-LSTM performed poorly because these methods only considered temporal correlation and not spatial correlation. LSTNet combined recurrent and CNNs to capture both temporal and spatial correlation, and therefore performed better compared to the previous methods. Graph WaveNet and MTGNN employ graph convolution to capture spatial correlation and therefore perform better on such unstructured data. Although both methods propose adaptive adjacency matrices, neither method is effective at making predictions without given graph adjacency matrices. On the New York City taxi dataset, our method improves the MAE and RMSE by 5.4% and 3.2%, respectively, over the recent method MTGNN. And on the San Francisco Bay Area bicycle data, our method improves the MAE and RMSE by 3.7% and 4.1%, respectively.

**TABLE 2** Comparison of different methods on the New York taxi and San Francisco Bay Area bicycle datasets.

| | NYC Taxi | | SF Bay Area Bike | |
|---|---|---|---|---|
| **Methods** | **MAE** | **RMSE** | **MAE** | **RMSE** |
| HA | 26.8689 | 40.2882 | 0.3995 | 0.8789 |
| SVR | 9.8898 | 14.9052 | 0.3908 | 0.7003 |
| XGBoost | 10.1720 | 15.8121 | 0.3087 | 0.6988 |
| FC-LSTM | 9.8061 | 14.9442 | 0.3378 | 0.7297 |
| LSTNet | 8.1639 | 13.0699 | 0.3065 | 0.7010 |
| Graph WareNet | 7.8026 | 12.5372 | 0.2865 | 0.6610 |
| MTGNN | 7.7729 | 12.2841 | 0.2779 | 0.6452 |
| **GSTGAT** | **7.3503** | **11.8867** | **0.2676** | **0.6189** |

GSTGAT, gated spatiotemporal graph attention network; HA, historical average; MAE, mean absolute error; RMSE, root mean square error; SVR, support vector regression.

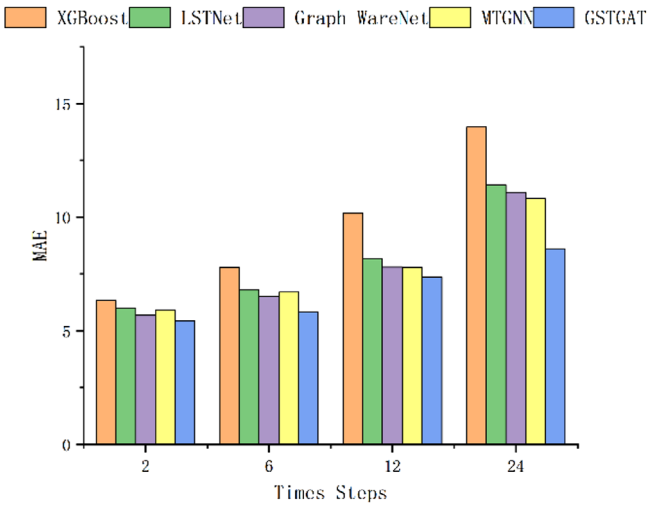
To validate the performance of our model on multi-step prediction, we conduct comparative experiments with four classical methods on the NYC Taxi and SF Bay Area Bike datasets. We predict 2 time steps, 6 time steps, 12 time steps and 24 time steps to compare the results. As shown in Figures 4 and 5, our model achieves the lowest MAE and RMSE. Due to the exclusive focus on temporal correlation, XGBoost performs the worst and the performance drops significantly when the prediction time interval increases. Graph WareNet and MTGNN, two spatiotemporal GCNs, perform better than LSTNet because they use graph convolution to capture spatial correlation. Similarly, because both methods use one-dimensional convolution to capture temporal correlation, they are less effective than our model and the performance gap in long-term prediction is more pronounced.

## 5.6 | Ablation studies

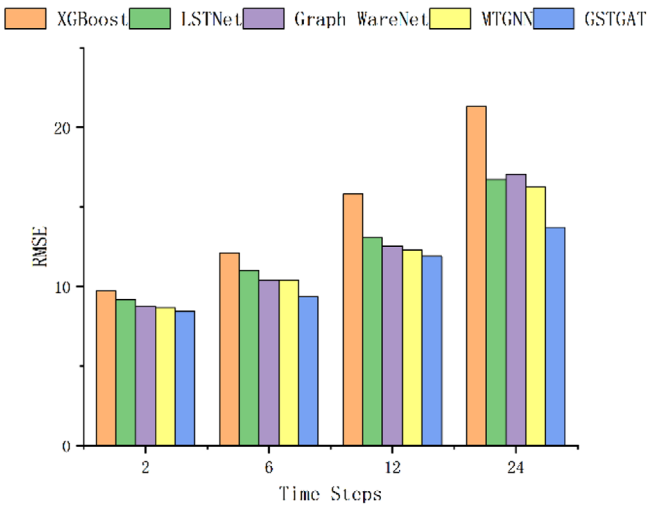
To validate the effectiveness of each module in the GSTGAT model, we conduct ablation experiments on the NYC Taxi and SF Bay Area Bike datasets to input 12 time steps of history to predict 12 time steps in the future. We remove different components from the GSTGAT model resulting in three different variants: (1) No Spatial: No spatial feature extraction module, (2) No GRGAU: No gating recurrent in the spatial feature extraction module, (3) No Gated: No gating in the temporal feature extraction module. The experimental results are shown in

[3] https://gitee.com/yaoshuilin/dataset
[4] https://pytorch.org

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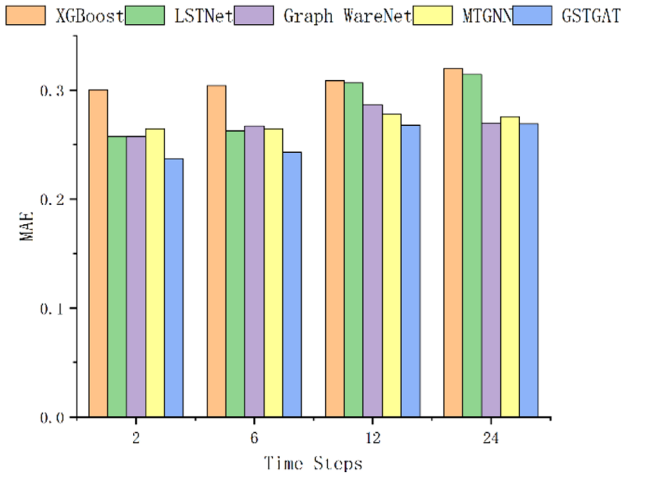



(a) MAE of NYC Taxi

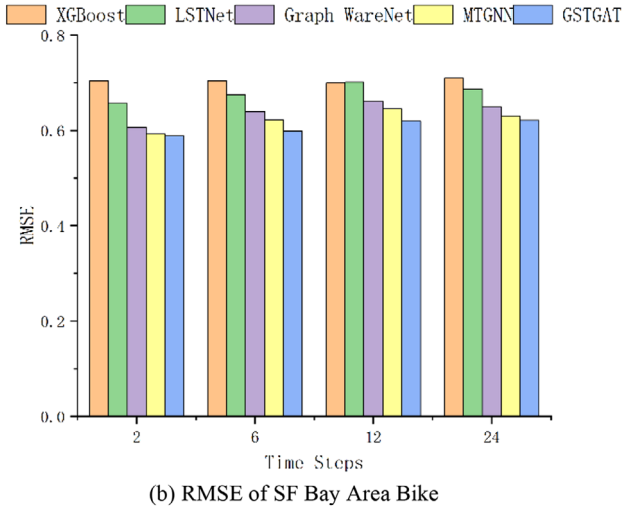



(b) RMSE of NYC Taxi

**FIGURE 4** Comparison of predictions at different time steps on the NYC Taxi dataset. (a) MAE of NYC Taxi, (b) RMSE of NYC Taxi. MAE, mean absolute error; RMSE, root mean square error; SVR, support vector regression.




(a) MAE of SF Bay Area Bike




(b) RMSE of SF Bay Area Bike

**FIGURE 5** Comparison of predictions at different time steps on the SF bay area bike dataset. (a) MAE of SF bay area bike, (b) RMSE of SF bay area bike. MAE, mean absolute error; RMSE, root mean square error.

Table 3. The inclusion of the spatial feature extraction module substantially improves the model performance, with MAE and RMSE increasing by 10.5% and 6.6%, respectively, in the NYC Taxi dataset, with MAE and RMSE increasing by 15.2% and 5.4%, respectively, in the SF Bay Area Bike dataset, as it enables the model to extract spatial correlations. The inclusion of gated recurrent between spatial feature distributions at different time steps and the addition of gating operations on the top of self-attention also improve model performance. Because it extracts dynamic information across spatial dimensions and reduces the effect of weak correlation information, which makes MAE and RMSE improve by 4.9% and 1%, respectively in the NYC Taxi dataset and improve by 3.2% and 4.6%, respectively, in the SF Bay Area Bike dataset with the addition of gated recurrent operation, and improve by 1.1% and 0.5%, respectively, in the NYC Taxi dataset and improve by 0.9% and 1.7%, respectively, with the addition of gating operation. By integrating these components, our model achieves the best performance.

**TABLE 3** Ablation experiments on the NYC taxi and SF bay area bike datasets.

| Methods | NYC Taxi | | SF Bay Area Bike | |
|---|---|---|---|---|
| | MAE | RMSE | MAE | RMSE |
| No gated | 7.4356 | 11.9469 | 0.2701 | 0.6293 |
| No GRGAU | 7.7329 | 12.0051 | 0.2763 | 0.6487 |
| No spatial | 8.2158 | 12.7296 | 0.3156 | 0.6544 |
| **GSTGAT** | **7.3503** | **11.8867** | **0.2676** | **0.6189** |

GRGAU, gated recurrent graph attention unit; GSTGAT, gated spatiotemporal graph attention network; MAE, mean absolute error; RMSE, root mean square error.

## 6 | CONCLUSION AND FUTURE WORK

In this paper, we propose an end-to-end multivariate spatiotemporal forecasting model applied to traffic demand forecasting. In GSTGAT, we employ a self-attention mechanism with gating

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for temporal feature extraction from traffic data, and propose a new approach to dynamically modelling traffic data in the spatial dimension by using graph attention combined with GRUs, and effectively capture spatiotemporal dependencies by combining them with the Transformer architecture. GSTGAT achieves excellent results on experiments conducted on two real-world available traffic datasets, with results showing that our model outperforms other baselines. A limitation of the work in this paper is that it does not take into account information on weather, temperature and wind speed, which are external factors that have a significant impact on traffic travel. We will investigate scalable solutions for applying GSTGAT on large-scale datasets and use more useful information to design better predictive models in future work.

## AUTHOR CONTRIBUTIONS

**Shuilin Yao**: Methodology; software; formal analysis; investigation; writing—original draft. **Huizhen Zhang**: Conceptualozation; supervision; writing—review & editing; funding acquisition. **Chenxi Wang**: Data curation; validation. **Dan Zeng**: Visualization. **Ming Ye**: Writing—review & editing.

## CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

## DATA AVAILABILITY STATEMENT

Data openly available in a public repository. The data that support the findings of this study are openly available in gitee at https://gitee.com/yaoshuilin/dataset.

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